

FARM TO HOME

Complete Technical Architecture, Supply Chain & Operational Blueprint (A to Z)

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Architecture: Direct Farm-to-Consumer (F2C)

GitHub Repository:
<https://github.com/mscharan6303/Farm-to-home.git>

Tech Stack: React, Node.js, Express, MongoDB, Socket.io

Document Version: 1.0.0 (Production Release)

Core Model: Demand-Driven Daily Harvest & 90% Net Farmer Payout

1. Executive Summary & Vision

Farm to Home is an end-to-end Direct Farm-to-Consumer (F2C) e-commerce ecosystem designed to revolutionize agricultural distribution in India. Traditional supply chains involve up to 4 to 6 layers of intermediaries (wholesalers, mandis, commission agents, regional distributors, and neighborhood retailers). This legacy model results in severe profit exploitation—where farmers earn less than 25-30% of consumer payments—and high post-harvest wastage due to extended transit times and prolonged warehouse cold storage.

Farm to Home replaces intermediary exploitation with a **transparent, technology-driven direct model**. The platform guarantees that **90% of gross produce sales are paid directly to farmers**, while retaining a modest 10% platform commission fee to sustain technology and logistics operations.

■ **Core Value Proposition:** Direct connectivity between farmers and consumers. Zero warehouse stock hoarding, guaranteed 90% net earnings for local farmers, and morning-harvested produce delivered straight to consumer doorsteps.

2. Produce Freshness Guarantee & Supply Chain Logistics

A primary concern in fresh produce distribution is ensuring absolute freshness without relying on artificial preservatives or long-term cold storage. Farm to Home operates on a **Demand-Driven Harvest Model** rather than an inventory-hoarding retail model.

A. How Freshness is Guaranteed (Harvest-on-Demand)

- **Zero Warehouse Stocking:** Unlike traditional supermarkets that store vegetables in distribution hubs for days, Farm to Home maintains zero inventory in central warehouses.

- **Morning Harvest Execution:** Produce listed on the platform is harvested by farmers early in the morning based on confirmed incoming consumer orders.
- **Direct Hub Aggregation:** Harvested packages are brought to local fulfillment hubs by 7:00 AM, where delivery agents immediately pick them up for doorstep delivery.
- **Harvest Availability Toggle:** Farmers possess live control via their dashboard to toggle operational status (█ Harvesting Active` vs `█ Harvesting Paused`). If adverse weather or field replenishment occurs, harvesting is paused instantly, preventing stale orders.

B. Managing Multi-Day Produce Shelf Life (e.g., Tomatoes, Brinjals)

Certain produce items like Tomatoes, Brinjals, Onions, and Potatoes naturally remain fresh for 4 to 7 days post-harvest. Questions often arise regarding how large quantities are handled when harvested in bulk:

1. **Batch Harvesting per Customer Order Volume:** Farmers do not harvest tons of produce speculatively. They monitor real-time order volumes on their dashboard and pick only the quantity required to fulfill confirmed orders.
2. **Field Preservation:** Crops remain unharvested on the plant until an order is locked, which preserves peak nutrient density and natural freshness superior to post-harvest cold storage.
3. **Multi-Day Listing & Stock Caps:** When farmers list items like Tomatoes, they set precise available stock caps (`kg`). As stock sells out, the status automatically updates, prompting fresh harvest scheduling.

C. Step-by-Step Delivery Logistics Workflow

Stage	Operational Action	Stakeholder Involved
1. Order Placement	Customer selects fresh produce, applies discounts/FarmPass, and places order online.	Consumer
2. Demand Aggregation	Orders are automatically grouped by farmer and displayed on the Farmer Dashboard.	System Engine
3. Morning Harvest	Farmer inspects demand, harvests fresh produce at dawn, and packages items.	Local Farmer
4. Hub Drop-off	Farmer drops off packaged orders at the local delivery hub by morning.	Farmer & Hub Coordinator
5. Task Lock & Pickup	Delivery Agent reviews orders in Agent Panel, checks Accept Task (locks order exclusively), and loads items.	Delivery Agent
6. Doorstep Delivery	Agent delivers fresh items, collects COD cash if applicable, verifies payment (`Paid █`), and marks order `Delivered █`.	Delivery Agent & Customer

3. System Architecture & Technology Stack

Farm to Home is built using a modern, scalable full-stack JavaScript architecture utilizing the MERN stack (MongoDB, Express, React, Node.js) paired with WebSocket real-time communication.

Layer / Subsystem	Technology Chosen	Technical Purpose & Functionality
Frontend UI Core	React 18 + Vite	High-performance single page application (SPA) with lightning-fast modular component rendering.
State Management	React Context API	Global state management for authentication tokens (`AuthContext`) and dynamic cart items (`CartContext`).
Styling & Icons	Custom CSS3 + React Icons	Fully mobile-responsive, modern CSS design without external heavy CSS frame overhead.
Backend Server API	Node.js + Express.js	RESTful API handling routing, middleware authentication, payment simulation, and database CRUD operations.
Database Storage	MongoDB + Mongoose ODM	Flexible document store mapping Users, Products, Orders, Video Ads, and Subscriptions with strict schemas.
Real-Time Messaging	Socket.io (WebSockets)	Direct live chat channel connecting customers and farmers for instant produce inquiry.
Multi-Tab State Sync	CloudSync Event Listeners	Synchronizes state (cart updates, order status changes) seamlessly across multiple open browser tabs.
Media Infrastructure	Cloudinary CDN + HTML5	Optimized storage and streaming for high-definition produce images and farmer promotional video ads.

4. Complete Module & Dashboard Breakdown (The 4 Core Panels)

A. ■ Customer Dashboard & Shopping Storefront

- **Promotional Video Ads Carousel:** Displays farmer video ads at the top of the homepage. Features interactive ■ *Add to Cart* and ■ *Order Now* overlay buttons. If no video ads are active, the container auto-hides for a 100% clean UI.
- **Category Navigation & Produce Catalog:** Allows instant filtering across *Vegetables*, *Fruits*, *Dairy*, *Grains*, and *Organic Specials* with search and pricing filters.
- **Dynamic Checkout & FarmPass Integration:** Calculates item subtotal, applies 10% FarmPass discount if subscribed, handles delivery address input, and supports COD or Online Payments.
- **Real-Time Order Tracking & Live Chat:** Customers track order fulfillment progress step-by-step and can initiate live Socket.io chat with the farmer.

B. ■ Farmer Dashboard & Seller Portal

- **Harvest Availability Switch:** Master switch allowing farmers to set `■ Harvesting Active` or `■ Harvesting Paused` status.
- **Net 90% Financial Summary:** Live metrics displaying Total Orders, Gross Sales, 10% Platform Fee, and **Net 90% Take-Home Earnings**.

- **Produce Catalog Management (CRUD):** Add new crops, edit prices/stock, upload images, and set organic tags.
- **Promotional Video Ad Creator:** Farmers can publish video ads for their produce, mapping the video to a specific item and selecting duration (1, 3, 7, 15, or 30 Days).

C. ■ Delivery Agent Panel & Logistics Hub

- **Hub Package Overview:** View morning produce packages dropped off by local farmers.
- **Single-Agent Order Lock:** Delivery drivers review available orders and click **Accept Task**. This instantly locks the task to that specific agent, preventing double-assignment.
- **COD Payment Verification:** Drivers collect cash on delivery, mark payment status as `Paid ■`, and update order delivery state.
- **Customer Contact Transparency:** Clear display of customer name, phone number, and delivery address.

D. ■ Master Platform Admin Control Center (5 Master Tabs)

- **Tab 1: Financials & Payouts:** Master platform revenue metrics (GMV, 10% Platform Fee, 90% Net Farmer Payouts). Features ****Clickable Farmer Financial Statement Modals**** detailing delivered produce items, total sales, 90% farmer profit, 10% platform profit, and a ****Paid ■ / Unpaid ■ Payout Selector****.
- **Tab 2: Product Catalog Control:** Full master CRUD (Add, Edit, Delete) across all listed produce.
- **Tab 3: Orders Master Control:** Complete order governance, status overrides, COD payment overrides, order deletion, and ****Assigned Delivery Agent Details****.
- **Tab 4: User & Accounts Directory:** Full control over Customer, Farmer, Delivery Agent, and Admin accounts with activate/suspend options.
- **Tab 5: Video Ads Control:** Admin governance over all farmer promotional video ads, including creation, approval, and deletion.

5. Platform Financial Architecture & Monetization

The financial model is engineered to prioritize farmer prosperity while maintaining self-sustaining platform economics.

Financial Component	Percentage / Amount	Operational Allocation & Purpose
Net Farmer Payout	90.0% of Gross Sales	Transferred directly to the producing farmer as take-home net profit.
Platform Commission	10.0% of Gross Sales	Retained by the platform owner to cover cloud hosting, server API maintenance, and payment processing fees.
Standard Shipping Fee	■49 per order	Charged on non-premium orders under ■300; paid to delivery agents for fulfillment.
FarmPass Pass Revenue	■499 / month	100% platform revenue stream providing consumers with free delivery and 10% extra discounts.

6. FarmPass Subscription Engine & Account Persistence

FarmPass Premium is the platform's flagship customer loyalty subscription program priced at **₹499/month**. Subscribers enjoy **Unlimited Free Delivery** on all orders and a **Flat 10% Extra Discount** applied automatically during checkout.

- **■ Online Payment Gateway Modal:** Customers can subscribe via a sleek online modal supporting simulated instant UPI transfers (``farmtohome@upi``) or Credit/Debit Card processing.
- **■ Permanent Account-Level Persistence:** Premium membership is tied directly to the user's account email (``premium_sub_``). When a user logs out and logs back in, their FarmPass Premium badge and discount benefits are permanently restored without loss of status.

7. Local Setup, Execution & Repository Structure

The project is structured into standard ``backend`` and ``frontend`` directories for seamless local development and production deployment.

Step	Command Execution	Notes / Description
1. Clone Repo	<code>`git clone https://github.com/mscharan6303/Farm-to-home.git`</code>	Clone official master repository.
2. Backend Setup	<code>`cd backend && npm install && npm run dev`</code>	Starts Express server on port 5000.
3. Frontend Setup	<code>`cd frontend && npm install && npm run dev`</code>	Launches Vite dev server at <code>http://localhost:5173</code> .
4. Build Verification	<code>`cd frontend && npm run build`</code>	Compiles production distribution bundle cleanly.

8. Conclusion & Project Sign-Off

Farm to Home stands as a robust, fully implemented Direct Farm-to-Consumer e-commerce platform. By uniting demand-driven harvesting, real-time logistics locking, farmer video advertising, 90% net farmer payout transparency, and multi-panel admin governance, the application bridges the gap between rural farmers and urban households efficiently and ethically.

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